Sell the harvest. Connect the whole journey.

A focused Sri Lankan solution for Google Cloud AI Builder Cup. Proposed theme: Sustainability & Social Impact.

The story: A farmer in Dambulla has tomatoes ready. A shop buyer in Colombo needs a larger order. A driver can move a suitable load, and a local worker can sort and load it. SaviPola connects the steps that turn those separate needs into a completed sale.

This is a scenario to validate with local users, not a claim that interviews or a field pilot have already been completed.

Person	Practical need	SaviPola response
Farmer	Describe and sell a harvest without complex forms	Gemini drafts a listing; farmer reviews it
Buyer	Understand volume, asking prices and fulfilment	Stock-aware sourcing proposal and AI explanation
Driver	Find a load appropriate for the vehicle	Capacity-checked booking and separate estimate
Local worker	Find paid sorting or loading work	Service listing, booking and completed-job review

Why the AI belongs here

Language to action: Gemini on Vertex AI extracts a farmer message into validated fields, preserving missing details as unknowns. **Photo to information:** it describes visible cosmetic condition and uncertainty. **Plan to understanding:** it explains an allocation computed from actual application stock.

One complete demo journey

Describe harvest / review listing / inspect visible condition / propose sourcing / negotiate and reserve / book transport and local help / fulfil and review.

The application calculates quantities and prices, and records stock changes. AI assists information handling; seller acceptance and human review remain explicit.

Build evidence for the rubric.

The official weights below guide our priorities. The evidence mapping is our recommendation, not a promise of selection.

Weight	What to prove	Show judges
40% technical + Gen AI	A functioning system with meaningful AI	Real Vertex outputs; stock reservation; deployed URL; persistent records
25% problem + impact	A specific, credible local need	One harvest story; early interviews if available; measurable pilot plan
25% innovation	Coordination that makes the transaction easier	Language-to-listing, photo observations, sourcing, logistics and local work
10% experience	People can finish the task	Clear kg/LKR labels, review steps, shortages and failures shown

Google services with a purpose

Service	Role in this implementation
Vertex AI + Gemini	Listing draft, provisional photo review, computed-plan explanation
Cloud Run	Host the FastAPI service and web interface
Cloud SQL PostgreSQL	Persist stock, offers, jobs and AI audit events across instances
Secret Manager	Supply the database password at deployment

Current status: integration source, Dockerfile and deployment script are included. Your Google Cloud project must still be configured and deployed. Successful live Vertex access is not claimed. A project-configured badge is separate from successful AI audit events.

Choose useful scope

Firebase Auth, Maps Routes, consented Cloud Storage evidence and BigQuery analytics are sensible later additions. They are not integrated in this version. A larger Google tool list does not substitute for a complete working transaction.

The configurable model default is gemini-3.5-flash-lite, listed in the Google lifecycle documentation when checked. Confirm project access before recording.

Make value visible. Measure it in the field.

Keep farmer proceeds and buyer landed cost separate. Do not describe sample arithmetic as established savings.

Illustrative 600 kg example	LKR
500 kg at 255/kg + 100 kg at 245/kg	152,000 goods value
210 km sample truck estimate: 1,800 + 65/km	15,450 freight
One sample loading job	3,800 labour
Goods + freight + labour	171,250 total / 285.42 per kg

All figures are examples, not live market rates or net farmer income. A two-farm pickup needs operational validation; the prototype does not optimize pooled routes. HARTI bulletins are a candidate source, but no live price feed is integrated.

A small, credible pilot

Proposed design: two weeks in one sourcing area, around ten farmers, two buyers, two drivers and five workers. These are planning counts, not enrolled participants. Start with consented interviews about a recent real transaction.

Measure	Compare against
Coordination minutes per fulfilled order	Comparable pre-pilot orders
Fulfilled kg / listed kg; cancellations and stock conflicts	Same crop/grade and harvest conditions
Buyer landed LKR/kg; farmer proceeds less selling costs	Comparable routes and independently recorded costs
Paid jobs, repeat bookings and disputes	Actual completed work and follow-up

A 30% reduction in coordination time is an initial unmeasured target. Publish the actual baseline, sample size and observed result after the pilot. Simulated trades cannot prove waste reduction or income gains.

Be precise about quality and trust

Photo opinions do not verify pesticides, pathogens, internal condition or safety. Grades are self-reported. Worker histories and identities shown in the demo are sample data. Each farmer separately accepts offers; grouped checkout and pooled freight routing remain future work.

FINISH AND SUBMIT

One focused demo. One clear submission.

Target a 2 minute 50 second English walkthrough. Deploy first, then record the real working flow.

Time	Scene
0:00-0:20	Introduce the farmer, buyer, driver and loader
0:20-0:50	Run a real Vertex listing draft; review and publish
0:50-1:10	Show photo observations and uncertainty
1:10-1:35	Show sourcing split and Gemini explanation
1:35-2:05	Negotiate, accept, reserve stock and book transport
2:05-2:25	Complete local work, review it and fulfil the trade
2:25-2:50	Show Google Cloud architecture and honest pilot measures

Split the remaining work

Builder A: deploy Cloud Run + Cloud SQL, verify real Vertex calls, and check persistence after redeployment. **Builder B:** refine the local story, prepare the English PDF slide deck and record the demo. Pair for the final full transaction.

Submit: working deployed prototype URL, public repository, video shorter than three minutes and a PDF deck. This document is a supporting brief, not a slide deck. The public timeline lists October 18, 2026; verify the exact cutoff time in the event dashboard.

Eligibility: the public rules require a 2-4 person JAPAC team aged 21+ and exclude current students. Check your team against the organizer rules.

Primary sources checked 17 October 2026

Organizer rubric and requirements

Organizer FAQ and submission rules

Event timeline

Google Gen AI SDK

Model lifecycle

Cloud Run and PostgreSQL